

Peptidoglycan Precursor Biosynthesis and Lipid II Export: A Mechanistic Review

A commissioned review-style synthesis. Scope: the reusable bacterial module that converts UDP-N-acetylglucosamine (UDP-GlcNAc) to UDP-MurNAc-pentapeptide, transfers it to the undecaprenyl carrier to make lipid I, glycosylates lipid I to lipid II, and flips lipid II across the cytoplasmic membrane. The module ends at lipid II export and excludes glycan polymerization, cross-linking, carrier recycling, and wall remodeling.

1. Executive Summary

Peptidoglycan (PG) precursor biosynthesis is a linear, largely universal bacterial assembly line that builds the membrane-anchored disaccharide-pentapeptide monomer, **lipid II**, and delivers it to the outer face of the cytoplasmic membrane for polymerization. The pathway partitions cleanly into three physical regimes: (i) a **soluble cytoplasmic phase** in which six enzymes (the enolpyruvyl transferase MurA, the flavin-dependent reductase MurB, and the four ATP-dependent amide ligases MurC–MurF), together with a convergent D-Ala–D-Ala branch (Ddl), elaborate the nucleotide-sugar UDP-MurNAc-pentapeptide; (ii) a **membrane-embedded biosynthetic phase** in which MraY transfers the phospho-MurNAc-pentapeptide unit to the polyprenyl carrier undecaprenyl phosphate (C55-P) to form lipid I, and MurG adds GlcNAc to form lipid II; and (iii) a **translocation phase** in which the flippase MurJ moves lipid II to the periplasmic leaflet.

The cytoplasmic chemistry is mechanistically well understood and enzymologically conserved; its enzymes have no eukaryotic counterparts and are validated or aspirational antibiotic targets (fosfomycin→MurA; D-cycloserine→Ddl/Alr; nucleoside natural products such as muraymycins/tunicamycin→MraY). The membrane and translocation steps are more contested. The identity of the lipid II flippase was, for over a decade, one of the most disputed questions in bacterial cell biology, with **FtsW**, **MurJ**, and **AmJ** all proposed before genetic and biochemical trapping experiments established **MurJ (a MOP-superfamily transporter)** as the primary flippase acting by an alternating-access cycle (PMIDs 26792999, 30482840, 35660157, 32129990) — though not the universal sole translocase, since *B. subtilis* carries an unrelated, stress-inducible backup flippase, Amj

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A recurring theme is that much "lipid II" biology is inferred from a handful of model organisms (chiefly *E. coli*, *B. subtilis*, *S. aureus*, *M. tuberculosis*) and should not be over-generalized.

2. Definition and Biological Boundaries

2.1 What is included

The module is the "Mur pathway plus the two lipid transferases plus the flippase":

Step	Enzyme (PSEPK canonical)	Reaction	Compartment
1	MurA	UDP-GlcNAc + PEP → enolpyruvyl-UDP-GlcNAc	cytoplasm
2	MurB	(NADPH-dependent reduction) → UDP-MurNAc	cytoplasm
3	MurC	+ L-Ala → UDP-MurNAc-L-Ala	cytoplasm
4	MurD	+ D-Glu → UDP-MurNAc-L-Ala-D-Glu	cytoplasm
5	MurE	+ <i>meso</i> -DAP (or L-Lys) → UDP-MurNAc-tripeptide	cytoplasm
(conv.)	Ddl (DdlB exemplar)	2 D-Ala → D-Ala-D-Ala	cytoplasm
7	MurF	+ D-Ala-D-Ala → UDP-MurNAc- pentapeptide	cytoplasm
8	MraY	+ C55-P → lipid I (+ UMP release)	inner leaflet
9	MurG	+ UDP-GlcNAc → lipid II	inner leaflet
10	MurJ	flip lipid II → outer leaflet	transmembrane

The pathway is **reusable**: the same lipid II is consumed by both the elongation (elongasome) and division (divisome) machineries. It is a metabolic converging funnel — all downstream wall growth depends on flux through this single set of enzymes.

2.2 Neighboring processes often conflated but excluded here

- **Glycan polymerization / transglycosylation** (aPBPs and SEDS proteins RodA/FtsW) — begins *after* export. Notably, the discovery that **RodA and FtsW are themselves PG glycosyltransferases** (SEDS family) reorganized the downstream field

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this is downstream of our boundary but historically entangled with the flippase debate because FtsW was itself a flippase candidate.

- **Transpeptidation / cross-linking** (DD- and LD-transpeptidases, PBPs) — extracytoplasmic.
- **Undecaprenyl-phosphate recycling** (UppP/BacA, UppS synthesis; C55-PP dephosphorylation after transglycosylation) — the carrier-regeneration loop that *feeds* MraY but is not part of monomer assembly.
- **UDP-GlcNAc supply** (GlmS/GlmM/GlmU) — the shared amino-sugar node feeding both PG and LPS/teichoic acid; it is the *substrate source*, upstream of MurA.
- **Stem-peptide tailoring** (e.g., MurT/GatD amidation of D-Glu to D-iso-Gln in *S. aureus*/mycobacteria; FemABX branch-peptide addition; DAP amidation) — lineage-specific decorations layered onto the core precursor.
- **Cell-wall remodeling / autolysins / recycling (Amp, Mpl)** — turnover, not synthesis.

2.3 Competing definitions

Some reviews draw the cytoplasmic boundary at MurF (the "Mur pathway" *sensu stricto*), treating MraY/MurG/MurJ as a separate "membrane cycle." Others fold MurJ into the polymerization machinery because it is functionally coupled to the divisome/elongasome. The brief's boundary (through lipid II export) is defensible and increasingly standard: it captures a self-contained "make and deliver the monomer" unit and stops before the combinatorial complexity of wall growth.

4. Major Molecular Players and Active Assemblies

4.1 Cytoplasmic phase

- **MurA (UDP-GlcNAc enolpyruvyl transferase).** Transfers the enolpyruvyl group of PEP to the 3'-OH of UDP-GlcNAc. Two-domain enzyme with a large inter-domain loop closure; Cys115 (E. coli numbering) forms the covalent adduct targeted by the epoxide antibiotic **fosfomycin**

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Notably, structural/kinetic analysis of the C115S mutant showed Cys115 is essential for **product release rather than the enolpyruvyl-transfer chemistry itself**, correcting an earlier assumption; consistently, **Cys115→Asp** substitutions confer intrinsic fosfomycin resistance in *Mycobacterium* and *Chlamydia*

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Numerous covalent-warhead inhibitors beyond fosfomycin have since been designed against Cys115

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MurA is the committed first step and a flux- control point. Mechanistically and structurally it is one of only two known enolpyruvyl transferases, the other being **EPSP synthase** of the shikimate pathway (fosfomycin/ glyphosate parallel) — a deep evolutionary kinship discussed in §5

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- **MurB (UDP-MurNAc dehydrogenase).** FAD-dependent, NADPH-driven reduction of the enolpyruvyl to D-lactyl, yielding UDP-MurNAc. Completes formation of the muramic-acid scaffold.
- **MurC/D/E/F (ATP-dependent Mur ligases).** A structurally and mechanistically related family (three-domain fold, central ATP-binding Rossmann-like domain) adding L-Ala, D-Glu, meso-DAP/L-Lys, and D-Ala-D-Ala respectively. All require divalent cations and share the acyl-phosphate mechanism

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They are enzymologically tractable as a reconstituted "one-pot" cascade

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and are inhibited by rationally designed and natural-product inhibitors (e.g., MurD D-Glu-mimetics,

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In some taxa (mycobacteria, corynebacteria) the ligases are regulated by **Ser/Thr protein-kinase phosphorylation** (PknA/PknB), linking wall synthesis to signaling

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- **Ddl (D-Ala-D-Ala ligase) + Alr (alanine racemase)**. ATP-dependent condensation of two D-Ala into the terminal dipeptide, a convergent input to MurF. **D-cycloserine** inhibits both Alr and Ddl. Ddl specificity is exquisitely tunable: a single active-site residue (Phe vs Tyr) switches output between the depsipeptide D-Ala-D-Lac and the dipeptide D-Ala-D-Ala

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the molecular basis of **VanA-type vancomycin resistance**

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and observed natively in glycopeptide-producing actinomycetes

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4.2 Membrane biosynthetic phase

- **MraY (phospho-MurNAc-pentapeptide transferase)**. Polytopic integral membrane enzyme of the polyprenyl-phosphate transferase (PNPT) superfamily. Transfers the phospho- MurNAc-pentapeptide moiety onto C55-P, releasing UMP and forming lipid I. Crystal structures (e.g., *Aquifex aeolicus* MraY, alone and with the natural inhibitor muraymycin D2) show a cytoplasmic active site with essential acidic residues and a Mg²⁺ cofactor, and large ligand-induced conformational changes

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Kinetics support ternary- complex catalysis with a conserved catalytic histidine

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MraY is the target of several nucleoside natural-product antibiotic classes (muraymycins, tunicamycin, liposidomycins/caprazamycins) (PMIDs 29386462, 33607457).

- **MurG (UDP-GlcNAc:lipid I β -1,4-N-acetylglucosaminyltransferase).** Membrane-associated GT-B-fold glycosyltransferase that adds GlcNAc to lipid I to complete lipid II. It follows a **compulsory ordered Bi Bi mechanism** (donor UDP-GlcNAc binds first) and, revealingly, does *not* require the full C55 chain for catalysis in vitro — the carrier lipid is important for pathway context, not for the chemistry

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Higher-order oligomeric/functional-unit organization has been proposed from crystallographic work

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4.3 Translocation phase

- **MurJ (lipid II flippase).** A **MOP (Multidrug/Oligosaccharidyl-lipid/Polysaccharide) superfamily** transporter with an inner amphipathic cavity. Multiple crystal structures capture **inward-facing, outward-facing, and "squeezed"** conformations, and MD plus in-cell cross-linking support an **alternating-access** cycle in which conserved cavity residues (e.g., Arg residues; A29, D269) bind the lipid II head group and drive the conformational transition (PMIDs 30482840, 35660157, 32129990, 29461535). Substrate recognition appears **coupled to conformational opening** — engineered MOP paralogs (WzxC) can be rewired to flip PG precursors, arguing that specificity is encoded in which substrate destabilizes the inward-open state

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Endogenous lipids (cardiolipin) and antibiotics (ramoplanin, vancomycin) modulate lipid II occupancy of MurJ

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4.4 Higher-order organization

The cytoplasmic enzymes are encoded largely within the conserved **dcw (division and cell wall) gene cluster** and co-transcribed with division genes (ftsQ/ftsW), physically and transcriptionally linking precursor supply to the cell cycle

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Protein–protein interaction and structural studies suggest the Mur enzymes and their

membrane partners are spatially organized rather than freely diffusing, plausibly channeling intermediates toward the divisome/elongasome (P 27136593).

5. Evolutionary and Cell-Biological Variation

Deep conservation. The Mur pathway is one of the most broadly conserved biosynthetic routes in the Bacteria; the cytoplasmic ligases share a common fold consistent with ancestral gene duplication and functional divergence (a canonical example of paralog expansion: MurC/D/E/F are best understood as descendants of a single ATP-dependent amide-ligase ancestor, with **MurC/MurD the most tractable representatives of the ancestral single-amino-acid-adding activity**). The entry step reaches even deeper: **MurA is one of only two known enolpyruvyl transferases**, sharing a two-domain architecture and PEP chemistry with **EPSP synthase** of the aromatic-amino-acid (shikimate) pathway — both enzymes are found essentially only in plants and microbes and absent from animals, and both are exploited agrochemically/therapeutically (fosfomycin vs glyphosate) (P 8994972).

This roots the pathway's first committed step in an ancient PEP-utilizing enzyme family predating the peptidoglycan-specific role. The undecaprenyl-phosphate carrier logic (nucleotide-sugar → lipid-linked oligosaccharide → flip → polymerize) is shared with **eukaryotic N-glycosylation and O-antigen/capsule biogenesis** — a deep, convergent or homologous "lipid-carrier" strategy for building glycoconjugates across a membrane (P 26792999).

MraY likewise belongs to a broadly distributed polyprenyl-phosphate transferase superfamily (with WecA and eukaryotic GPT relatives), indicating the membrane-transfer chemistry is itself ancient.

Lineage variation. - *Position-3 residue:* meso-DAP (Gram-negatives, bacilli, mycobacteria/actinobacteria) vs L-Lys (many Gram-positive cocci, e.g., *S. aureus*, streptococci) — a MurE substrate-specificity difference (P 18194336).

- *Stem-peptide tailoring:* amidation of D-Glu (MurT/GatD) and of DAP; interpeptide bridges (e.g.,

pentaglycine via FemABX in *S. aureus*) — added on the UDP or lipid-linked precursor and highly lineage-specific (

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- *Terminal dipeptide plasticity*: D-Ala-D-Ala → D-Ala-D-Lac/D-Ala-D-Ser in glycopeptide-resistant enterococci/staphylococci and glycopeptide producers (PMIDs 10529248, 9435111). - *MraY/MurG homologs*: MraY belongs to a superfamily including WecA (which primes O-antigen with GlcNAc-1-P) and WbpL/GPT — the transferase step is ancient and paralogously expanded; the **PG-specific MraY** is the best representative for the ancestral peptidoglycan role.

Cell-biological / physiological state variation. The *same* lipid II pool supplies **elongation** (MreB-guided elongasome) and **division** (FtsZ-guided divisome). Flux and localization of MurG and MurJ are spatially patterned with the cell cycle; MurJ activity at the division site is regulated within the divisome. Sporulating and dormant states (spores, *M. tuberculosis* persistence), L-forms (transient wall-less states), and endosymbionts with reduced genomes (e.g., *Arsenophonus*, from which a MurJ structure was solved,

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represent physiological/compartmental variants — with **cell-wall-less lineages (Mollicutes, Chlamydia's reduced wall, most Archaea) representing lineage-specific loss or replacement** (archaea use pseudomurein or other envelopes and lack the canonical Mur pathway).

Alternative routes to the same outcome. Beyond the Van ligase depsipeptide bypass, the key example within our boundary is the flippase question itself: **different organisms and assays implicated different proteins (FtsW/Amj/MurJ)** for the same translocation outcome. The most striking demonstration is that *B. subtilis* cells lacking **all ten MOP-superfamily members (including MurJ)** remain viable, because they encode an unrelated backup flippase, **Amj (alternate to MurJ; ydaH)**, that is synthetic-lethal with MurJ and, like MurJ, can support lipid II flipping and viability when expressed in *E. coli* lacking its own MurJ

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Amj is induced by the envelope-stress σ^M regulon, so loss or inhibition of MurJ can trigger a compensatory translocase. MOP paralogs can also be *engineered* to substitute

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Together these show the transport function is **not uniquely tied to one protein family across all lineages** — an important caveat for MurJ-targeted drugs and for extrapolating *E. coli* flippase biology to all bacteria.

6. Constraints, Dependencies, and Failure Modes

Mandatory ordering (why it cannot be reshuffled). - Each Mur ligase is a **product-templated recognition** enzyme: MurD requires the MurC product, MurE the MurD product, MurF the MurE product. The stepwise amide bonds cannot be formed out of order because the enzyme active sites read the growing UDP-MurNAc-peptide. - **MraY must follow MurF:** it transfers a *pentapeptide*-bearing phospho-MurNAc; the nucleotide precursor must be complete first. (Some tailoring, e.g., FemABX bridges, is the exception — it occurs on lipid I/II after MraY.) - **MurG must follow MraY:** MurG glycosylates lipid I, not the soluble precursor, and binds donor sugar first (ordered Bi Bi)

([P 12022887](#)).

- **MurJ must follow MurG:** only the complete disaccharide-pentapeptide (lipid II) is the physiological flippase substrate.

Compartment / topology constraints. - Lipid I and lipid II are **synthesized on the inner (cytoplasmic) leaflet**; polymerization occurs on the **outer leaflet** — hence an obligatory flip

([P 26792999](#)).

This topological inversion is the *raison d'être* of MurJ and the reason the flippase step is non-negotiable. - The **carrier lipid is limiting**. C55-P is present in low copy and shared with LPS, teichoic acid, and capsule pathways; sequestration of lipid II (e.g., by nisin, vancomycin, ramoplanin, or a stalled flippase) is lethal because it **starves the recycling pool**, a major failure mode exploited by antibiotics.

Mutually exclusive / substrate-specific events. - Ddl's D-Ala-D-Ala vs D-Ala-D-Lac outputs are mutually exclusive per enzyme variant and determine vancomycin susceptibility (PMIDs 29686137, 10529248). - MurE's DAP vs Lys specificity is fixed per organism.

Evidence ruling out otherwise plausible paths. - In-cell cross-linking trapped lipid II *on* MurJ and showed Arg mutants bind but fail to transport — ruling out a purely passive/spontaneous flip and supporting an active, residue-dependent conformational cycle

([P 32129990](#)).

- The "squeezed" MurJ conformation (cavity collapsed) argues against a static open channel and for alternating access

([P 35660157](#)).

- MurG's indifference to the full C55 chain in vitro rules out the hypothesis that the long isoprenoid is chemically required for glycosyl transfer (P 12022887).

Failure modes / therapeutic vulnerabilities. Fosfomycin (MraA... MurA), D-cycloserine (Alr/Ddl), tunicamycin/muraymycins/caprazamycins (MraY), and lipid-II-sequestering glycopeptides/lantibiotics (vancomycin, nisin, ramoplanin) all act within or immediately at the boundary of this module; MurB, the Mur ligases, and MurJ are actively pursued but lack clinically approved inhibitors — a standing opportunity given their essentiality and absence of human homologs (PMIDs 27734910, 27136593).

7. Controversies and Open Questions

1. **Flippase identity and mechanism.** Although MurJ is now accepted as the primary lipid II flippase, the historical FtsW/MurJ/AmJ dispute

(P 26792999)

reflected genuinely conflicting evidence: purified **FtsW was shown biochemically to induce transbilayer movement of lipid II** in model membranes and *E. coli* vesicles

(P 21386816),

whereas **MurJ** carried the essential in-vivo flippase phenotype and the conformational-trapping data (PMIDs 30482840, 32129990). This is a textbook in-vitro-vs-in-vivo discordance rather than a fully reconciled model. Compounding this, MurJ is **not the universal sole flippase**: *B. subtilis* uses the unrelated, stress-inducible **Amj** as a redundant translocase

(P 25918422),

so the "flippase" is lineage-dependent rather than a single conserved protein. Open: Does **FtsW retain any flippase-like or "hand-off" role** now that it is established as a SEDS glycosyltransferase

(P 35274942)?

How many bacteria carry Amj-like backups, and are they induced clinically when MurJ is

inhibited? How is MurJ energized — is the conformational cycle coupled to the membrane potential/ion gradients, and if so, which ion?

2. **Where does the flip physically hand off to polymerization?** The spatial/temporal coupling of MurJ to RodA/FtsW-PBP complexes at the divisome/elongasome is not resolved at the structural level

([P 27136593](#)).

3. **Metabolic channeling vs free diffusion.** Whether Mur enzymes form a bona fide multienzyme complex that channels intermediates, or simply co-localize via the *dcw* operon and membrane recruitment, remains debated

([P 23555903](#)

[P 27136593](#)).

4. **Regulation.** Ser/Thr-kinase phosphorylation of Mur ligases is documented in actinobacteria

([P 23555903](#)),

but the systemic logic coupling precursor flux to growth rate, nutrient status, and the cell cycle is incompletely mapped, and may not generalize across phyla.

5. **Organism-comparability caveat.** Much mechanism is stitched together from *E. coli* (MurJ genetics), *Aquifex/Bacillus* (MraY structure/kinetics), *S. aureus* (tailoring), and *M. tuberculosis* (ligase regulation). Cross-organism extrapolation is a standing source of over-generalization; claims should be qualified by the source system.

6. **Structural gaps.** High-resolution structures of MraY and MurG *with intact lipid-linked substrates*, and of MurJ *caught mid-flip with lipid II*, remain incomplete — the central barrier to a fully atomistic model of the membrane phase.

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Search date: 2026-07-23 (PubMed). This review synthesizes primary literature and authoritative reviews; it is explanatory rather than encyclopedic and flags organism-comparability limits throughout.

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